

Assignment-1.4

“GPU and Cooling system.”

TASK-1

Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

My laptop has an Intel Iris Xe Graphics GPU. It is an integrated GPU, which means the graphics processor is built into the same processor package as the CPU rather than being a separate graphics card.

Brand: Intel

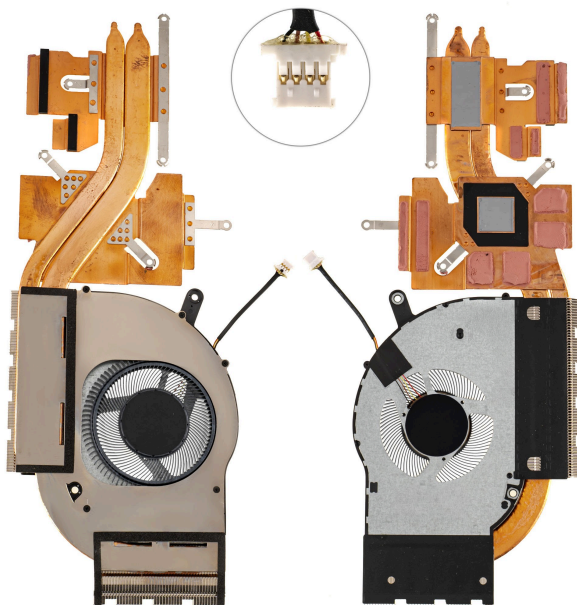
Model: Intel Iris Xe Graphics

Type: Integrated GPU

The GPU is responsible for processing and displaying graphics, videos, animations, and images on the laptop screen. It can also help with graphics-intensive applications and games.

TASK-2

Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.



1. Copper Cold Plate / Thermal Block:

- **Function:** Sits directly on top of the CPU/GPU die using a thin layer of thermal paste to eliminate air gaps.
- **Mechanism:** It absorbs thermal energy directly from the semiconductor die through conduction.

2. Copper Heat Pipe(s):

- **Function:** Transports heat rapidly from the heat source (processor) to the heatsink fins located near the exhaust vents.
- **Mechanism:** Contains a small amount of liquid working fluid under low pressure inside a hollow copper tube with a porous wick structure. The liquid boils at the hot end, turns into vapor, travels to the cooler end near the fan, condenses back into liquid, and travels back through capillary action.

3. Aluminum / Copper Heatsink (Fin Array):

- **Function:** Maximizes the surface area available for heat transfer.
- **Mechanism:** As the heat pipe transfers thermal energy to these ultra-thin metallic fins, heat is dissipated into the air passing through them.

4. Centrifugal Blower Fan:

- **Function:** Forces continuous airflow across the heatsink fins to expel trapped heat outside the chassis.
- **Mechanism:** Pulls cooler ambient air through intake vents located on the bottom of the laptop chassis and blows hot exhaust air out through the side or rear ventilation grilles.

TASK-3

Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Comparison Criteria	Air Cooling Systems (Heatsink & Fans)	Liquid Cooling Systems (AIO / Custom Loop)
1. Upfront Cost & Value	Inexpensive: Basic air coolers cost between \$25 and \$90. Standard option for budget and mid-range PC builds.	Expensive: All-in-One (AIO) coolers range from \$80 to \$250, while custom liquid loops can exceed \$500+.
2. Cooling Efficiency & Performance	Moderate: Efficient for everyday tasks and moderate gaming, but ambient room temperature heavily limits maximum cooling performance during prolonged heavy loads.	Superior: Liquid transfers thermal energy away from the GPU die significantly faster than air, keeping temperatures lower during continuous heavy workloads.
3. Maintenance, Reliability & Risk	Zero Maintenance & Safe: Requires minimal maintenance beyond clearing dust from fan blades. No risk of fluid leakage or mechanical pump failures.	Requires Monitoring & Risk: Mechanical pumps can wear out or fail over time. Minor risk of fluid leaks or air bubbles getting trapped in the loop.
4. Noise Levels	Higher under heavy load: Fans must spin at higher RPMs to keep high-end GPUs cool under stress, resulting in louder fan noise.	Quieter overall: Larger radiators allow fans to spin at lower speeds while maintaining lower thermal levels.

5. Space & Case Clearance	Bulky Heatsinks: Large tower coolers require wide PC cases and can obstruct motherboard RAM slots.	Sleek CPU/GPU Block: The block mounted on the chip is small, but requires mounting space for radiators (240mm–360mm) along case walls.
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TASK-4

Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

To handle simultaneous high-definition streaming (like broadcasting live IPL matches) and playing graphics-heavy AAA games, I choose an **All-in-One (AIO) Liquid Cooling system** for the GPU setup.

Simultaneous 3D game rendering and real-time video encoding (streaming) push a dedicated graphics card to 100% utilization for hours at a time, generating intense thermal output. Liquid cooling provides a much higher thermal heat capacity than traditional air cooling, absorbing sudden temperature spikes instantly and preventing the GPU from thermal throttling (slowing down clock speeds to protect itself from overheating). Furthermore, maintaining lower operating temperatures prolongs the operational lifespan of high-end silicon components. Finally, liquid cooling operates at significantly lower noise levels, ensuring that loud cooling fans do not bleed unwanted background noise into the live stream's microphone audio during intense match broadcasts.